

PROBLEM FRAMING MEMO

-#- -P-R-O-B-L-E-M- -F-R-A-M-I-N-G- -M-E-M-O-
 -*-*T-o-:-*-* -K-P-C- -I-n-u-k-a- -F-o-u-n-d-a-t-i-o-n- -B-o-a-r-d- -&- -C-
 o-m-p-l-i-a-n-c-e- -C-o-m-m-i-t-t-e-e-
 -*-*F-r-o-m-:-*-* -S-y-n-e-r-g-y- -H-a-c-k-a-t-h-o-n- -D-a-t-a- -G-o-v-e-r-
 n-a-n-c-e- -T-e-a-m-
 -*-*D-a-t-e-:-*-* -A-u-g-u-s-t- -2-1-, -2-0-2-6-
 -*-*S-u-b-j-e-c-t-:-*-* -B-e-n-e-f-i-c-i-a-r-y- -P-r-i-v-a-c-y-, -C-o-n-s-e-
 n-t- -C-o-m-p-l-i-a-n-c-e-, -&- -K-D-P-A- -D-a-t-a- -G-o-v-e-r-n-a-n-c-e- -
 F-r-a-m-e-w-o-r-k-

-#-#- -E-x-e-c-u-t-i-v-e- -S-u-m-m-a-r-y-
 -T-h-e- -r-a-p-i-d- -d-i-g-i-t-i-z-a-t-i-o-n- -o-f- -b-e-n-e-f-i-c-i-a-r-y- -p-r-
 o-f-i-l-e-s- -a-c-r-o-s-s- -t-h-e- -f-o-u-r- -p-i-l-l-a-r-s- -o-f- -t-h-e- -*-*K-
 P-C- -I-n-u-k-a- -F-o-u-n-d-a-t-i-o-n-*-* -(S-c-h-o-l-a-r-s-h-i-p-, -P-l-u-s-
 -, -V-o-c-a-t-i-o-n-a-l-, -T-e-c-h-) -e-x-p-o-s-e-s- -t-h-e- -f-o-u-n-d-a-t-i-
 o-n- -t-o- -s-e-v-e-r-e- -o-p-e-r-a-t-i-o-n-a-l-, -f-i-n-a-n-c-i-a-l-, -a-n-d-
 -l-e-g-a-l- -l-i-a-b-i-l-i-t-i-e-s-. -U-n-d-e-r- -t-h-e- -*-*K-e-n-y-a- -D-a-t-
 a- -P-r-o-t-e-c-t-i-o-n- -A-c-t- -(K-D-P-A)-*-*-, -n-o-n--c-o-m-p-l-i-a-n-c-
 e- -w-i-t-h- -d-a-t-a- -p-r-o-c-e-s-s-i-n-g- -p-e-r-m-i-s-s-i-o-n-s- -c-a-r-r-i-
 -e-s- -s-t-a-t-u-t-o-r-y- -p-e-n-a-l-t-i-e-s- -o-f- -u-p- -t-o- -*-*K-E-S- -5- -
 M-i-l-l-i-o-n-*-* -o-r- -1%- -o-f- -a-n-n-u-a-l- -t-u-r-n-o-v-e-r-, -a-l-o-n-
 g-s-i-d-e- -s-e-v-e-r-e- -r-e-p-u-t-a-t-i-o-n-a-l- -r-i-s-k-s.-

-W-i-t-h- -1-, -0-0-0- -a-c-t-i-v-e- -b-e-n-e-f-i-c-i-a-r-y- -r-e-c-o-r-d-s-, -d-
 -a-t-a- -i-s- -a-c-c-e-s-s-e-d- -b-y- -m-u-l-t-i-p-l-e- -i-n-t-e-r-n-a-l- -a-c-t-
 o-r-s- -(H-Q- -D-i-r-e-c-t-o-r-s-, -P-i-l-l-a-r- -C-o-o-r-d-i-n-a-t-o-r-s-, -
 R-e-g-i-o-n-a-l- -F-i-e-l-d- -O-f-f-i-c-e-r-s-, -a-n-d- -A-u-d-i-t-o-r-s)-. -
 S-h-a-r-i-n-g- -u-n-m-a-s-k-e-d- -d-a-t-a-s-e-t-s- -c-o-n-t-a-i-n-i-n-g- -n-a-
 m-e-s-, -N-a-t-i-o-n-a-l- -I-D-s-, -p-h-o-n-e- -n-u-m-b-e-r-s-, -a-n-d- -e-
 m-a-i-l-s- -v-i-o-l-a-t-e-s- -c-o-r-e- -K-D-P-A- -p-r-i-n-c-i-p-l-e-s.-

-T-h-i-s- -m-e-m-o- -f-r-a-m-e-s- -t-h-e- -p-r-o-b-l-e-m- -b-o-u-n-d-a-r-i-e-
 s-, -q-u-a-n-t-i-f-i-e-s- -t-h-e- -c-o-m-p-l-i-a-n-c-e- -r-i-s-k-s-, -a-n-d- -o-
 -u-t-l-i-n-e-s- -o-u-r- -d-e-p-l-o-y-e-d- -t-e-c-h-n-i-c-a-l- -s-o-l-u-t-i-o-n-s-
 -f-o-r- -t-h-e- -I-n-u-k-a- -P-o-r-t-a-l.-

-#-#- -T-h-e- -C-h-a-l-l-e-n-g-e-:- -P-r-i-v-a-c-y-, -C-o-n-s-e-n-t-, -&- -I-
 n-g-e-s-t-i-o-n- -Q-u-a-l-i-t-y-

-#-#-#- -1.- -T-h-e- -P-l-l- -E-x-p-o-s-u-r-e- -B-o-t-t-l-e-n-e-c-k-
 -*-*T-h-e- -R-i-s-k-*-*:- -R-a-w- -c-o-n-t-a-c-t- -d-e-t-a-i-l-s- -(N-a-
 m-e-s-, -N-a-t-i-o-n-a-l- -I-D-s-, -p-h-o-n-e- -n-u-m-b-e-r-s-, -e-m-a-i-l-
 s)- -a-r-e- -e-x-p-o-s-e-d- -t-o- -v-a-r-i-o-u-s- -c-o-o-r-d-i-n-a-t-o-r-s.- -
 E-x-t-e-r-n-a-l- -r-e-p-o-r-t- -d-i-s-s-e-m-i-n-a-t-i-o-n- -a-n-d- -u-n-a-u-t-h-
 -o-r-i-z-e-d- -i-n-t-e-r-n-a-l- -a-c-c-e-s-s- -p-a-t-t-e-r-n-s- -i-n-c-r-e-a-s-e-
 - -t-h-e- -p-r-o-b-a-b-i-l-i-t-y- -o-f- -d-a-t-a- -b-r-e-a-c-h-e-s.-
 -*-*T-h-e- -S-o-l-u-t-i-o-n-*-*:- -S-e-r-v-e-r---s-i-d-e- -d-y-n-a-m-i-c-
 -C-o-l-u-m-n---L-e-v-e-l- -S-e-c-u-r-i-t-y- -(C-L-S)- -t-h-a-t- -a-u-t-o-m-a-

-t-i-c-a-l-l-y- -m-a-s-k-s- -p-e-r-s-o-n-a-l- -d-a-t-a- -f-o-r- -n-o-n---a-u-t-h
-o-r-i-z-e-d- -r-o-l-e-s- -(l-i-k-e- -A-u-d-i-t-o-r-s-),- -c-o-m-b-i-n-e-d- -w
-i-t-h- -R-o-w---L-e-v-e-l- -S-e-c-u-r-i-t-y- -(R-L-S)- -r-e-s-t-r-i-c-t-i-n-g
- -r-e-g-i-o-n-a-l- -o-f-f-i-c-e-r-s- -t-o- -t-h-e-i-r- -s-p-e-c-i-f-i-c- -d-e-p-o
-t-s-.-

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-#-#-#- -2-.- -l-n-g-e-s-t-i-o-n- -Q-u-a-l-i-t-y- -C-o-n-t-r-o-l- -(G-r-e-a-t-  
-E-x-p-e-c-t-a-t-i-o-n-s- -G-a-t-e-s)-
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-R-a-w- -f-i-e-l-d- -r-e-g-i-s-t-r-a-t-i-o-n- -d-a-t-a- -o-f-t-e-n- -c-o-n-t-a-i-  
n-s- -m-i-s-s-i-n-g- -f-i-e-l-d-s,- -i-n-c-o-r-r-e-c-t- -r-e-g-i-o-n-s,- -o-r- -  
i-n-v-a-l-i-d- -p-h-o-n-e- -n-u-m-b-e-r-s.- -M-a-n-u-a-l- -d-a-t-a- -s-c-r-u-  
b-b-i-n-g- -w-a-s-t-e-s- -h-o-u-r-s- -o-f- -a-d-m-i-n-i-s-t-r-a-t-i-v-e- -t-i-m-  
e:-
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-*- *-*-T-h-e- -R-i-s-k-*-*-:- -C-o-r-r-u-p-t-e-d- -d-a-t-a-b-a-s-e-s- -l-e-a-
d- -t-o- -a-u-d-i-t- -v-a-l-i-d-a-t-i-o-n- -f-a-i-l-u-r-e-s- -a-n-d- -m-i-s-d-i-r-
e-c-t-e-d- -s-t-i-p-e-n-d- -p-a-y-o-u-t-s-.-

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-*- *-*-T-h-e- -S-o-l-u-t-i-o-n-*-:- -l-n---m-e-m-o-r-y- -G-r-e-a-t- -E-x-p-
e-c-t-a-t-i-o-n-s- -v-a-l-i-d-a-t-i-o-n- -s-u-i-t-e-s- -t-h-a-t- -i-n-t-e-r-c-e-p-
-t- -s-t-r-e-a-m-e-d- -r-e-g-i-s-t-r-a-t-i-o-n- -e-v-e-n-t-s- -a-n-d- -q-u-a-r-a-
-n-t-i-n-e- -n-o-n---c-o-m-p-l-i-a-n-t- -p-a-y-l-o-a-d-s- -a-u-t-o-m-a-t-i-c-a-
l-l-y.-
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-#-#-#- -3-. -G-r-a-n-u-l-a-r- -C-o-n-s-e-n-t- -P-u-r-p-o-s-e-s- -M-a-n-a-g-  
e-m-e-n-t-
```

-*- *- *-T-h-e- -P-o-l-i-c-y- *- *-:- -U-n-d-e-r- -t-h-e- -K-D-P-A,- -b-e-n-e-f-
i-c-i-a-r-i-e-s- -m-u-s-t- -h-a-v-e- -t-h-e- -r-i-g-h-t- -t-o- -c-h-o-o-s-e- -h-
o-w- -t-h-e-i-r- -d-a-t-a- -i-s- -p-r-o-c-e-s-s-e-d- (-S-t-i-p-e-n-d- -s-h-a-r-
i-n-g,- -m-e-d-i-a- -p-h-o-t-o-s,- -S-M-S- -n-o-t-i-f-i-c-a-t-i-o-n-s,- -a-n-
d- -e-x-t-e-r-n-a-l- -d-o-n-o-r- -r-e-p-o-r-t-i-n-g)-.-

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-*- -T-h-e- -R-i-s-k-*-:- -A- -f-e-l-l-o-w- -m-u-s-t- -b-e- -a-b-l-e- -t-o-
- -w-i-t-h-d-r-a-w- -c-o-n-s-e-n-t- -o-r- -r-e-q-u-e-s-t- -p-r-o-f-i-l-e- -e-r-a
-s-u-r-e- -("-R-i-g-h-t- -t-o- -b-e- -F-o-r-g-o-t-t-e-n-"- -u-n-d-e-r- -S-e-c-t
-i-o-n- -4-0-)- -a-t- -a-n-y- -t-i-m-e-. -M-a-n-u-a-l-l-y- -p-r-o-c-e-s-s-i-n-g
- -t-h-e-s-e- -r-e-q-u-e-s-t-s- -t-a-k-e-s- -u-p- -t-o- -*--1-4- -b-u-s-i-n-e-s
-s- -d-a-y-s-*- -p-e-r- -r-e-q-u-e-s-t-, -c-a-u-s-i-n-g- -m-a-s-s-i-v-e- -c-
o-m-p-l-i-a-n-c-e- -l-a-t-e-n-c-y.-
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-*- -*-T-h-e- -S-o-l-u-t-i-o-n-*-:- -L-i-v-e- -s-e-l-f--s-e-r-v-i-c-e- -d-e-
 s-k- -w-h-e-r-e- -b-e-n-e-f-i-c-i-a-r-i-e-s- -t-o-g-g-l-e- -p-r-e-f-e-r-e-n-c-e-
 s- -i-n-s-t-a-n-t-l-y,- -a-n-d- -a- -c-h-a-t-b-o-t- -r-i-g-h-t-s- -e-n-g-i-n-e-
 -t-h-a-t- -h-a-n-d-l-e-s- -r-e-q-u-e-s-t-s- -i-n- -S-w-a-h-i-l-i- -a-n-d- -E-n-
 g-l-i-s-h-.-

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-I-N-U-K-A- -D-A-T-A- -I-N-G-E-S-T-I-O-N- -&- -C-O-N-S-E-N-T- -M-A-N-A-
G-E-M-E-N-T- -P-I-P-E-L-I-N-E-

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-[-F-i-e-l-d- -E-v-e-n-t-]- --->- [-G-E- -Q-u-a-l-i-t-y- -G-a-t-e-]- --->- [-C-l-e-a-n- -D-a-t-a-b-a-s-e-]- --->- [-D-y-n-a-m-i-c- -R-L-S-/-C-L-S- -Q-u-e-r-y-]-
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-----V-----V-----
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-(Q-u-a-r-a-n-t-i-n-e)- - - - -(-R-e-j-e-c-t- -B-a-d-)- - - - -
 - - - - -(-O-b-f-u-s-c-a-t-e- -P-l-l-)-
 --- -M-a-s-k- -N-a-m-e-s-,- -E-m-a-i-l-s-
 --- -E-x-c-l-u-d-e- -R-e-g-i-o-n-
 -+-----
 -----+
 -\`-\`-\`-

-#-#- -P-r-o-p-o-s-e-d- -T-e-c-h-n-i-c-a-l- -l-n-t-e-r-v-e-n-t-i-o-n-s-
 -W-e- -h-a-v-e- -s-u-c-c-e-s-s-f-u-l-l-y- -d-e-p-l-o-y-e-d- -t-h-e- -f-o-l-l-o-
 w-i-n-g- -m-o-d-u-l-e-s- -t-o- -a-d-d-r-e-s-s- -t-h-e-s-e- -g-a-p-s:-
 -1-.- -*-B-i-l-i-n-g-u-a-l- -C-o-m-p-l-i-a-n-c-e- -C-h-a-t-b-o-t- -R-i-g-h-t-
 s- -E-n-g-i-n-e-*-*:-
 --- -U-s-e-s- -N-L-P- -i-n-t-e-n-t- -c-l-a-s-s-i-f-i-e-r- -t-o- -p-r-o-c-e-s-s-
 -u-s-e-r- -r-e-q-u-e-s-t-s- -i-n- -E-n-g-l-i-s-h- -a-n-d- -S-w-a-h-i-l-i.-
 --- -D-y-n-a-m-i-c-a-l-l-y- -c-o-m-p-i-l-e-s- -r-e-a-l---t-i-m-e- -p-i-l-l-a-r- -
 a-n-d- -r-e-g-i-o-n-a-l- -c-o-m-p-l-i-a-n-c-e- -s-t-a-t-u-s- -r-e-p-o-r-t-s- -(
 e-.g.- -c-o-u-n-t-i-n-g- -t-o-t-a-l- -f-e-l-l-o-w-s,- -a-c-t-i-v-e- -c-o-n-s-e-
 -n-t- -r-a-t-e-s,- -a-n-d- -a-c-t-i-v-e- -p-r-o-f-i-l-e- -e-r-a-s-u-r-e-s)-.-
 -2-.- -*-E-x-p-o-r-t- -C-o-m-p-l-i-a-n-t- -D-o-n-o-r- -R-e-p-o-r-t- -T-o-o-l
 -*-:-
 --- -A- -p-r-o-m-i-n-e-n-t- -d-a-s-h-b-o-a-r-d- -b-u-t-t-o-n- -t-h-a-t- -e-x-p-
 o-r-t-s- -t-h-e- -a-c-t-i-v-e- -d-i-r-e-c-t-o-r-y- -v-i-e-w- -i-n-t-o- -a- -f-u-l-
 l-y- -a-n-o-n-y-m-i-z-e-d- -C-S-V,- -c-o-m-p-l-e-t-e-l-y- -r-e-d-a-c-t-i-n-g-
 -s-e-n-s-i-t-i-v-e- -P-l-l- -c-o-l-u-m-n-s- -f-o-r- -e-x-t-e-r-n-a-l- -d-o-n-o-r
 - -a-u-d-i-t-s.-
 -3-.- -*-G-r-e-a-t- -E-x-p-e-c-t-a-t-i-o-n-s- -l-n-g-e-s-t-i-o-n- -S-t-r-e-a-
 m-*-*:-
 --- -C-l-e-a-n-s-e-s,- -s-t-a-n-d-a-r-d-i-z-e-s,- -a-n-d- -v-a-l-i-d-a-t-e-s-
 -r-e-g-i-s-t-r-a-t-i-o-n-s- -i-n- -r-e-a-l- -t-i-m-e,- -r-o-u-t-i-n-g- -f-a-i-l-u-
 r-e-s- -t-o- -q-u-a-r-a-n-t-i-n-e- -l-o-g-s.-